

Lab program-9:

9. Illustrate the concept of inter-process communication using shared memory with a C program

```
lab_9.c > main()
1  #include <stdio.h>
2  #include <windows.h>
3  #include <string.h>
4
5  #define SHM_SIZE 1024
6
7  int main()
8  {
9      HANDLE hMapFile;
10     char *pBuf;
11
12     // Create a named shared memory object
13     hMapFile = CreateFileMapping(
14         INVALID_HANDLE_VALUE, // Use system paging file
15         NULL, // Default security
16         PAGE_READWRITE, // Read/Write access
17         0,
18         SHM_SIZE,
19         "MySharedMemory" // Name of shared memory
20     );
21
22     if (hMapFile == NULL)
23     {
24         printf("Could not create file mapping (%lu)\n", GetLastError());
25         return 1;
26     }
27
28     // Map the shared memory into the process address space
29     pBuf = (char *)MapViewOfFile(
30         hMapFile,
31         FILE_MAP_ALL_ACCESS,
32         0,
33         0,
34         SHM_SIZE
35     );
36
37     if (pBuf == NULL)
38     {
39         printf("Could not map view (%lu)\n", GetLastError());
40         CloseHandle(hMapFile);
41         return 1;
42     }
43
44     // Write data
45     strcpy(pBuf, "Hello, shared memory!");
46
47     printf("Data written to shared memory: %s\n", pBuf);
48
49     // Cleanup
50     UnmapViewOfFile(pBuf);
51     CloseHandle(hMapFile);
52
53     return 0;
54 }
```

OUTPUT:

```
● PS C:\Users\SMILE\Documents\OS\Lab programs> ./lab_9.exe  
Data written to shared memory: Hello, shared memory!
```